

Projection-operator approach to potential scattering

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The projection-operator approach of Feshbach is applied to potential scattering. The aim is to describe single-particle or shape resonances in a mathematically rigorous manner as discrete states interacting with a continuum, in analogy to the well-known description of closed-channel resonances in scattering from targets with internal degrees of freedom. A projection operator Q is defined as $Q = \sum_{\nu=1}^N |\chi_{\nu}\rangle\langle\chi_{\nu}|$, where $\{|\chi_{\nu}\rangle\}$ is an arbitrary orthonormal set of L^2 functions. The complementary P space is spanned by a set of scattering states obtained in explicit form by orthogonalizing the free continuum to the set $\{|\chi_{\nu}\rangle\}$. The free Green's function in P space is constructed explicitly and the P -space scattering problem is solved with the use of separable expansions of the potential. Two standard model problems— s -wave scattering from the square-well potential and the δ -shell potential—are solved exactly, with the use of an arbitrary number of eigenstates of a particle in a spherical box to define the Q space. It is shown that the formalism leads to a decomposition of the exact T matrix and scattering phase shift into an orthogonality scattering, a direct scattering, and a resonant scattering contribution. The pole structure of the corresponding S matrices in the complex momentum plane is analyzed. Finally, the question of how to construct the appropriate discrete state, which projects out a given resonance, is briefly discussed.

I. INTRODUCTION

The projection-operator formalism of Feshbach^{1,2} proved to be particularly fruitful for the description of resonant scattering and reactions in nuclear, atomic, and molecular physics. In its original formulation, the formalism is based on the introduction of projection operators P and Q which project onto open and closed channels, respectively. Resonances are then naturally described as closed-channel states embedded in and interacting with the continuum of open channels.^{1,2} The formalism has found widespread application in the calculation of energies and widths of nuclear fine-structure resonances³ and electronic autoionizing states of atomic and molecular systems.⁴

In addition to these closed-channel resonances there exist also single-particle or shape resonances which owe their existence to the temporary trapping of the incident particle behind a barrier of the effective (local) potential. The single-particle resonances are usually broader than the closed-channel resonances but, nevertheless, are of particular interest in molecular physics. Due to the anisotropy of the molecular charge distribution, shape resonances are a very common phenomenon in electron-molecule scattering.⁵ They play an extraordinary role in processes involving energy exchange between electrons and the nuclear motion of molecules. Typical processes of this type are vibrational excitation of molecules or molecular ions by electron impact, dissociative attachment, and dissociative recombination.⁶ These collision and reaction processes constitute a major part of the physics and chemistry in outer space, in the higher atmosphere, and in plasmas and discharges.

The Feshbach formalism was first applied to molecular problems by Chen⁷ and O'Malley.⁸ Since then this for-

malism has found widespread application in the description of nuclear dynamics in resonant electron-molecule scattering, mainly as a formal tool and as the basis for a phenomenological description of resonance effects. For example, the local complex potential model for the nuclear motion in resonance states⁹ can be derived within this formalism.^{7,8,10} In particular, the formalism lends itself to a comprehensive description of resonance and threshold effects. The ubiquitous long-range potentials in electron-atom and electron-molecule scattering lead to singularities in the T matrix at $k=0$. These singularities can be incorporated into the Feshbach formalism using a threshold expansion of the complex level-shift function. One thus obtains a unified description of both the resonance, bound-state and virtual-state poles of the T matrix as well as the threshold singularities of the T matrix caused by the long-range potentials.¹¹

Despite its obvious usefulness as a qualitative tool, the projection-operator formalism has found very little application as a quantitative computational tool for single-particle resonances. When attempting a rigorous application of the formalism to potential scattering, two conceptual and practical problems arise which are not present in the description of closed-channel resonances. These problems are the following.

(i) The requirement that the continuum states spanning the P space should be orthogonal to the discrete states spanning the Q space is a major obstacle in practical calculations. In addition, one has to avoid overcompleteness of the set of states spanning the P and Q spaces. A rigorous application of the formalism requires the solution of the direct scattering problem in P space. The P -space scattering states are needed to calculate the direct scattering amplitude and the width function $\Gamma(E)$. It was soon recognized that the solution of the scattering problem with

the additional orthogonality constraint is more complicated than the solution of the original problem.

(ii) The appropriate choice of the states spanning the Q space is not clear from the outset. Even for a narrow resonance the choice of a square-integrable wave function "representing" this resonance is not obvious. For broad above-barrier resonances the construction of a suitable projector becomes even more ambiguous.

To the knowledge of the author, these difficulties have hitherto prevented rigorous applications of the projection-operator formalism to potential scattering resonances. A notable exception are the recent calculations of Hazi and coworkers on molecular shape resonances.¹² Hazi¹³ recognized that one need not solve the scattering problem in P space with proper boundary conditions to obtain the width function $\Gamma(E)$. Rather, one can use Stieltjes imaging methods¹⁴ to construct a smooth function $\Gamma(E)$ from a discrete representation of the background scattering continuum. The level shift $\Delta(E)$ is then obtained as the Hilbert transform of $\Gamma(E)$. This approach turns out to be very powerful as it allows the immediate implementation of computational methods for bound states which have developed to a high degree of sophistication in atomic and molecular physics. The drawback of the method is that one cannot obtain the background scattering amplitude or phase shift. Therefore, this approach does not provide the complete solution of the scattering problem.

Nesbet¹⁵ has discussed the application of the Feshbach formalism to shape resonances on a more formal level. He gives exact formal expressions for the P -space scattering states and the resonant phase shift for the example of a spherically symmetric potential function. Since in his approach the full Green's function of the scattering problem is assumed to be given, the implementation of this method for electron-atom or electron-molecule scattering is probably difficult and has not been attempted so far. An interesting model application, where both resonant and background phase shifts are obtained, can be found in Ref. 16.

The main motivation for a rigorous realization of the projection-operator formalism for potential scattering is the separation of the total scattering amplitude or phase shift into a resonance and a background term. The separation should be such that the background phase shift becomes a smooth function of energy in the energy interval of interest. All rapid variations of the phase shift should be contained in the resonant term. Such a separation of the phase shift is of considerable practical importance when the dynamics of nuclear motion during the electron scattering process is considered. Electronic resonance energies calculated in the fixed-nuclei approximation usually depend strongly on the internuclear distance(s), causing a breakdown of the Born-Oppenheimer (BO) approximation for the full scattering wave function.¹⁷ If one succeeds in extracting the resonances and the threshold effects (for example, rapid variations of the phase shift caused by a virtual state close to threshold), the background scattering wave function can be calculated in the BO approximation. The resonant scattering, on the other hand, can be treated beyond the BO approximation in the Feshbach formalism. The non-BO effects are simply reflected by the nonlocality (dependence on the nuclear momentum operator) of the optical

potential governing the nuclear motion.^{7,8} The problem can also be viewed as the task of constructing appropriate "diabatic" electronic states embedded in the continuum.¹⁸ The term diabatic means that the electronic wave function Φ of this state depends sufficiently weakly on the internuclear distance(s) such that Φ commutes with the kinetic-energy operator of the nuclei to a good approximation. These diabatic electronic states play a distinguished role in the description of atomic collision processes¹⁹ and molecular vibronic spectra.²⁰ The ultimate goal of the present work is to extend the well-established methods for the construction of bound diabatic states into the continuum region.

The main result of this paper is the resolution of problem (i) mentioned above. It will be shown that the orthogonal continuum states and the Green's function describing free propagation in the orthogonal continuum can be constructed in closed form for an arbitrary set of discrete states. The formulas for the explicit orthogonalization of the free continuum to a set of bound states are not new; it seems that they have been discovered independently several times in nuclear and solid-state physics (see below). The construction of the orthogonal continuum opens the way for the explicit solution of the scattering problem in P space. Since the orthogonalization is equivalent to the scattering by a certain separable potential, the choice of separable approximations to the potential in P space suggests itself, rendering the solution of the P -space scattering problem only slightly more complicated than the direct solution of the full problem. The Schwinger variational principle²¹ appears to be particularly useful in this context.

The formalism will be applied to solve two standard model problems (s -wave scattering from the square-well potential and the δ -shell potential) exactly within the projection-operator formalism. As the Q space we choose the first N s -wave states of a particle in a box of radius R_0 . This choice of the projection operator is particularly simple and appealing, since in the limit $N \rightarrow \infty$ the projectors become projection operators in coordinate space: Q project onto the interior region $r < R_0$, P projects onto the exterior region $r > R_0$. It will be shown that the projection-operator formalism leads to a separation of the exact T -matrix element and phase shift into an orthogonality scattering term, a direct scattering term, and a resonant scattering term. The sum of the first two represents the background scattering. We will also investigate the poles of the individual T matrices in the complex momentum plane. It will be seen that the resonant T matrix possesses—for any number of discrete states—all (infinitely many) poles of the exact T matrix. All but a finite number of these poles are canceled by nearby zeros (which are the poles of the background T matrix). The analysis of the analytic structure of the projection-operator formalism for exactly soluble models should prove useful for the understanding of more complicated situations.

An important problem which will only be touched upon in Sec. V, but not finally resolved, is to give a general prescription for the choice of the discrete state(s) [problem (ii) mentioned above]. This choice is not clear, especially for very broad resonances. In contrast to the orthogonality problem (i), the proper choice of the discrete state(s) is

not a formal mathematical question, but rather requires physical intuition.

II. THE FORMALISM

A. General

The general framework of the projection-operator formalism has been introduced and comprehensively discussed by Feshbach^{1,2} and Fonda and Newton.²² Therefore, only a brief outline will be given here.

We consider the potential scattering problem described by the Hamiltonian

$$H = K + V, \quad (2.1)$$

where K is the kinetic-energy operator and V is an arbitrary local potential. As usual, we start by introducing the projection operators P and Q ,

$$P^2 = P, \quad Q^2 = Q, \quad P + Q = 1, \quad (2.2)$$

with the requirement

$$\langle \vec{r} | Q | \psi \rangle \rightarrow 0 \quad (2.3)$$

for any $|\psi\rangle$. In particular, we choose

$$Q = \sum_{n=1}^N |\chi_n\rangle \langle \chi_n|, \quad (2.4)$$

where the $\langle \vec{r} | \chi_n \rangle$ are square-integrable functions. For simplicity, we assume that the $\{|\chi_n\rangle\}$ form an orthonormal set

$$\langle \chi_n | \chi_{n'} \rangle = \delta_{nn'}. \quad (2.5)$$

The P space is spanned by states $|\hat{k}^{(\pm)}\rangle$ which are orthogonal to the $\{|\chi_n\rangle\}$ and taken to be energy normalized:

$$\langle \chi_n | \hat{k}^{(\pm)} \rangle = 0, \quad (2.6a)$$

$$\langle \hat{k}^{(\pm)} | \hat{k}'^{(\pm)} \rangle = \delta(k^2/2 - k'^2/2) \delta(\Omega_k - \Omega_{k'}). \quad (2.6b)$$

In terms of these states the projection operator P can be written as

$$P = \int k dk d\Omega_k |\hat{k}^{(\pm)}\rangle \langle \hat{k}^{(\pm)}|. \quad (2.7)$$

Here and in the following the caret indicates orthogonality to the set of discrete states $\{|\chi_n\rangle\}$ spanning the Q space.

Of course, the $|\hat{k}^{(\pm)}\rangle$ are not uniquely determined by the orthogonality constraint (2.6). We will need, more specifically, the eigenstates of K_{PP} defined as the solutions of the

$$T_0(\vec{k}', \vec{k}) = \langle \vec{k}' | (K_{PP} - K) | \hat{k}^{(+)} \rangle, \quad (2.17)$$

$$T_{\text{dir}}^{(F)}(\vec{k}', \vec{k}) = \langle \hat{k}'^{(-)} | V_{PP} | \hat{k}^{(+)} \rangle, \quad (2.18)$$

$$T_{\text{res}}^{(F)}(\vec{k}', \vec{k}) = \langle \hat{k}'^{(-)} | H_{PQ}(k^2/2 - H_{QQ} - H_{QP} \hat{G}^{(+)} H_{PQ})^{-1} H_{QP} | \hat{k}^{(+)} \rangle. \quad (2.19)$$

The T matrix is thus given as the sum of three terms, which we may call the orthogonality scattering (T_0), the direct scattering (T_{dir}), and the resonant scattering (T_{res}) transition matrices. The orthogonality scattering T matrix

Lippmann-Schwinger (LS) equation

$$|\hat{k}^{(\pm)}\rangle = |\vec{k}\rangle + G_0^{(\pm)}(K_{PP} - K) |\hat{k}^{(\pm)}\rangle, \quad (2.8)$$

where $|\vec{k}\rangle$ is the free continuum state (plane wave),

$$G_0^{(\pm)} = (k^2/2 - K \pm i\eta)^{-1} \quad (2.9)$$

in the free Green's operator (η is the usual positive infinitesimal) and $K_{PP} = PKP$. The solutions of Eq. (2.8) are constructed in explicit form below.

Given the projection operators P and Q , the Schrödinger equation can be reduced to a Schrödinger equation in P space with the effective Hamiltonian^{1,2}

$$\mathcal{H} = H_{PP} + H_{PQ}(k^2/2 - H_{QQ})^{-1} H_{QP}. \quad (2.10)$$

Following Feshbach,^{1,2} we obtain for the P -space part of the total wave function

$$\begin{aligned} P | \Psi \rangle &= |\hat{\psi}_{\vec{k}}^{(+)}\rangle \\ &+ \hat{G}^{(+)} H_{PQ}(k^2/2 - H_{QQ} - H_{QP} \hat{G}^{(+)} H_{PQ})^{-1} \\ &\times H_{QP} |\hat{\psi}_{\vec{k}}^{(+)}\rangle, \end{aligned} \quad (2.11)$$

where we have introduced the reduced resolvent

$$\hat{G}^{(+)} = P(k^2/2 - H_{PP} + i\eta)^{-1} P. \quad (2.12)$$

The $|\hat{\psi}_{\vec{k}}^{(\pm)}\rangle$ are the direct scattering states defined as the solutions of the LS equation

$$|\hat{\psi}_{\vec{k}}^{(\pm)}\rangle = |\vec{k}^{(\pm)}\rangle + \hat{G}_0^{(\pm)} V_{PP} |\hat{\psi}_{\vec{k}}^{(\pm)}\rangle, \quad (2.13)$$

where

$$\hat{G}_0^{(\pm)} = P(k^2/2 - K_{PP} \pm i\eta)^{-1} P. \quad (2.14)$$

Combining Eqs. (2.8), (2.11), and (2.13) we obtain

$$\begin{aligned} P | \Psi \rangle &= |\vec{k}\rangle + G_0^{(+)}(K_{PP} - K) |\hat{k}^{(+)}\rangle + \hat{G}_0^{(+)} V_{PP} |\hat{\psi}_{\vec{k}}^{(+)}\rangle \\ &+ \hat{G}^{(+)} H_{PQ}(k^2/2 - H_{QQ} - H_{QP} \hat{G}^{(+)} H_{PQ})^{-1} \\ &\times H_{QP} |\hat{\psi}_{\vec{k}}^{(+)}\rangle. \end{aligned} \quad (2.15)$$

The transition matrix $T(\vec{k}', \vec{k})$, giving the amplitude for a transition $\vec{k} \rightarrow \vec{k}'$ is (the superscript F stands for Feshbach)

$$T(\vec{k}', \vec{k}) = T_0(\vec{k}', \vec{k}) + T_{\text{dir}}^{(F)}(\vec{k}', \vec{k}) + T_{\text{res}}^{(F)}(\vec{k}', \vec{k}), \quad (2.16)$$

where

does not appear in the usual Feshbach projection-operator formalism where the projection is defined with respect to channels^{1,2,22} since orthogonality of the channel wave functions is guaranteed by the Hermiticity of the corre-

sponding channel Hamiltonian. In potential scattering, on the other hand, there is only one channel and one has to enforce the orthogonality of the P and Q spaces explicitly, which results in the appearance of the orthogonality scattering term in Eq. (2.16). For the ensuing discussion it is convenient to combine the orthogonality and the direct scattering T matrices to give the background scattering T matrix

$$T_{bg}^{(F)}(\vec{k}', \vec{k}) = T_0(\vec{k}', \vec{k}) + T_{dir}^{(F)}(\vec{k}', \vec{k}). \quad (2.20)$$

A central quantity in the Feshbach formalism is the complex level-shift matrix

$$F_{nm}(k) = \langle \chi_n | H_{QP} \hat{G}^{(+)}(k) H_{PQ} | \chi_m \rangle. \quad (2.21)$$

The poles of the resonant transition matrix $T_{res}^{(F)}(\vec{k}, \vec{k}')$ in the complex momentum plane are given by

$$\det[\frac{1}{2}k^2\delta_{nm} - H_{nm} - F_{nm}(k)] = 0, \quad (2.22)$$

where $H_{nm} = \langle \chi_n | H | \chi_m \rangle$. It is seen that the real eigenvalues of H_{QQ} are shifted into the complex (energy or momentum) plane by the level-shift matrix $F_{nm}(k)$, yielding the complex resonance poles of the T matrix.

The projection-operator formalism as outlined so far provides a very transparent and convenient scheme for describing resonances. It provides the basis for many phenomenological models and approximation methods. For example, the Breit-Wigner formula for isolated or overlapping resonances follows immediately.^{1,2} Threshold singularities caused by long-range potentials are easily incorporated by a threshold expansion of $F_{nm}(k)$.¹¹ The formalism provides also a powerful tool to treat the nuclear dynamics in electron-molecule scattering beyond the Born-Oppenheimer approximation.^{7,8} In these heuristic approaches, the explicit construction of the orthogonal continuum states $|\hat{k}^{(\pm)}\rangle$ and the calculation of the direct scattering states $|\hat{\psi}_{\vec{k}}^{(\pm)}\rangle$ is avoided, either by neglecting the orthogonality constraint and the direct scattering contributions, or by introducing suitable phenomenological parametrizations to account for these effects. To develop the projection-operator formalism into a rigorous and computationally feasible tool for potential scattering, we have to construct explicitly the proper orthogonalized continuum $|\hat{k}^{(\pm)}\rangle$ and to find methods to solve for the direct scattering states $|\hat{\psi}_{\vec{k}}^{(\pm)}\rangle$ and the corresponding Green's operator $\hat{G}^{(\pm)}$.

B. Orthogonality scattering

Methods for orthogonalizing the free continuum to a given set of bound states have been invented independently several times in nuclear and solid-state physics.²³⁻²⁶ The central idea is to introduce a Hermitian model Hamiltonian $\tilde{H}$ which possesses the given set $\{|\chi_n\rangle\}$ (and only this set) as bound states. It suffices then to construct the scattering states of the model Hamiltonian $\tilde{H}$ to obtain the desired orthogonalized continuum. This approach ensures automatically the internal orthogonality of the continuum [i.e., Eq. (2.6b)] and the completeness of the total set $\{|\chi_n\rangle, |\hat{k}^{(+)}\rangle\}$ of states.

The simplest choice for the model Hamiltonian is^{23,25}

$$\tilde{H} = \sum_{n=1}^N |\chi_n\rangle \langle \chi_n| + K_{PP} = Q + K_{PP}. \quad (2.23)$$

The scattering states of $\tilde{H}$ are obtained as solutions of the LS equation

$$|\hat{k}^{(\pm)}\rangle = |\vec{k}\rangle + G_0^{(\pm)}(\tilde{H} - K) |\hat{k}^{(\pm)}\rangle. \quad (2.24)$$

Note that Eq. (2.24) is identical with Eq. (2.8) since $Q |\hat{k}^{(\pm)}\rangle = 0$. Equation (2.24) can be rewritten as

$$|\hat{k}^{(\pm)}\rangle = |\vec{k}\rangle - G_0^{(\pm)} Q K |\hat{k}^{(\pm)}\rangle. \quad (2.25)$$

Since the kernel of Eq. (2.25) is separable, the solution is immediately obtained in closed form^{23,25}

$$|\hat{k}^{(\pm)}\rangle = |\vec{k}\rangle - \sum_{n,m=1}^N G_0^{(\pm)} |\chi_n\rangle (\underline{A}^{-1})_{nm} \langle \chi_m | \vec{k} \rangle, \quad (2.26)$$

where

$$A_{nm} = \langle \chi_n | G_0^{(\pm)} | \chi_m \rangle. \quad (2.27)$$

From Eq. (2.26) it is obvious that the $|\hat{k}^{(\pm)}\rangle$ are indeed orthogonal to the $|\chi_n\rangle$. The orthogonality scattering T matrix is now immediately obtained as

$$T_0(\vec{k}', \vec{k}) = - \sum_{n,m=1}^N \langle \vec{k}' | \chi_n \rangle (\underline{A}^{-1})_{nm} \langle \chi_m | \vec{k} \rangle. \quad (2.28)$$

The phase shift of the orthogonalized continuum states is described by the K matrix

$$K_0(\vec{k}', \vec{k}) = -\pi \sum_{n,m=1}^N \langle \vec{k}' | \chi_n \rangle (\tilde{\underline{A}}^{-1})_{nm} \langle \chi_m | \vec{k} \rangle, \quad (2.29)$$

where

$$\tilde{A}_{nm} = \langle \chi_n | G_0^{(P)} | \chi_m \rangle = \langle \chi_n | \text{Re}\{G_0^{(\pm)}\} | \chi_m \rangle \quad (2.30)$$

are the matrix elements of the free Green's function with standing-wave boundary conditions. Finally, the free resolvent in P space, defined in Eq. (2.14), is explicitly given by

$$\hat{G}_0^{(\pm)} = G_0^{(\pm)} - \sum_{n,m=1}^N G_0^{(\pm)} |\chi_n\rangle (\underline{A}^{-1})_{nm} \langle \chi_m | G_0^{(\pm)}. \quad (2.31)$$

C. The direct scattering problem

The orthogonality scattering problem has thus been solved in closed form. To calculate $T_{dir}^{(F)}(\vec{k}', \vec{k})$ and $T_{res}^{(F)}(\vec{k}', \vec{k})$ in Eqs. (2.18) and (2.19) we need the direct scattering states $|\hat{\psi}_{\vec{k}}^{(+)}\rangle$ and the full P -space resolvent operator $\hat{G}^{(+)}$. Having seen that orthogonalization of the continuum to the discrete states is equivalent to scattering by a separable potential, it is natural to continue using separable expansions of the potential V_{PP} to calculate $|\hat{\psi}_{\vec{k}}^{(+)}\rangle$ and $\hat{G}^{(+)}$. Let us consider the separable potential²⁷⁻³⁰

$$V_{PP}^{(S)} = \sum_{i,j=1}^M PV |\hat{\chi}_i\rangle \langle \underline{V}^{-1} \rangle_{ij} \langle \hat{\chi}_j | VP, \quad (2.32a)$$

where $\underline{V}$ is an $M \times M$ matrix with elements

$$V_{ij} = \langle \hat{\chi}_i | V | \hat{\chi}_j \rangle \quad (2.32b)$$

and $\{|\hat{\chi}_j\rangle\}$ is a set of square-integrable basis functions in P space, i.e.,

$$P |\hat{\chi}_j\rangle = |\hat{\chi}_j\rangle, \quad Q |\hat{\chi}_j\rangle = 0. \quad (2.33)$$

The $|\hat{\chi}_j\rangle$ are basis functions used to generate a separable expansion of V_{PP} and should not be confused with the states $|\chi_n\rangle$ spanning the Q space. Solving the LS equation for the separable potential (2.32a) leads to the following expressions for the direct T matrix [defined in Eq. (2.18)²⁷⁻³⁰]

$$T_{\text{dir}}^{(F)}(\vec{k}', \vec{k}) = \sum_{i,j=1}^M \langle \hat{\chi}_i | V | \hat{\chi}_j \rangle \langle \underline{M}^{-1} \rangle_{ij} \langle \hat{\chi}_j | V | \hat{\chi}_i \rangle, \quad (2.34a)$$

where

$$M_{ij} = \langle \hat{\chi}_i | (V - V\hat{G}_0^{(+)}V) | \hat{\chi}_j \rangle \quad (2.34b)$$

and $\hat{G}_0^{(+)}$ is given by Eq. (2.31). This T -matrix expression is also obtained from the Schwinger variational principle²¹

$$\begin{aligned} T_{\text{dir}}^{(F)}(\vec{k}', \vec{k}, [\hat{\psi}]) &= \langle \hat{\chi}_i | V | \hat{\psi}_{\vec{k}}^{(+)} \rangle + \langle \hat{\psi}_{\vec{k}}^{(-)} | V | \hat{\chi}_j \rangle \\ &\quad - \langle \hat{\psi}_{\vec{k}}^{(-)} | (V - V\hat{G}_0^{(+)}V) | \hat{\psi}_{\vec{k}}^{(+)} \rangle \end{aligned} \quad (2.35)$$

with linear variation of the trial functions $\hat{\psi}_{\vec{k}}^{(\pm)}$.^{28,30} The variational property of the T matrix (2.34) guarantees a fast convergence of the expansion in practical applications.^{31,32}

From Eq. (2.34) we obtain immediately the expression for the direct scattering states [defined in Eq. (2.13)]

$$\begin{aligned} |\hat{\psi}_{\vec{k}}^{(+)}\rangle &= |\hat{\chi}_i\rangle + \sum_{j=1}^M \hat{G}_0^{(+)}V |\hat{\chi}_j\rangle \langle \underline{M}^{-1} \rangle_{ij} \langle \hat{\chi}_j | V | \hat{\chi}_i \rangle \\ &\quad + \sum_{j=1}^M \hat{G}_0^{(+)}V |\hat{\chi}_j\rangle \langle \hat{\chi}_j | V | \hat{\chi}_i \rangle \end{aligned} \quad (2.36)$$

and the full resolvent operator in P space [defined in Eq. (2.12)]

$$\hat{G}^{(+)} = \hat{G}_0^{(+)} + \sum_{i,j=1}^M \hat{G}_0^{(+)}V |\hat{\chi}_i\rangle \langle \underline{M}^{-1} \rangle_{ij} \langle \hat{\chi}_j | V \hat{G}_0^{(+)} \quad (2.37)$$

With these expressions we can calculate the complex level-shift matrix $F_{nm}(k)$ defined in Eq. (2.21) and the resonant T matrix defined in Eq. (2.19).

As will be shown in the following sections, the above formalism provides a manageable scheme to perform rigorous projection-operator calculations for potential scattering. To evaluate the Schwinger-type expressions (2.34), (2.36), and (2.37) we need matrix elements of V ,

$\hat{G}_0^{(+)}$, and $V\hat{G}_0^{(+)}V$ with the L^2 basis states $|\hat{\chi}_i\rangle$ in P space as well as the bound-continuum matrix elements $\langle \hat{\chi}_i | V | \hat{\chi}_j \rangle$ and $\langle \hat{\chi}_i | \hat{G}_0^{(+)} | \hat{\chi}_j \rangle$. With the use of the explicit orthogonalization formulas (2.26) and (2.31), these matrix elements can be reduced to ordinary matrix elements of V , $G_0^{(+)}$, and $VG_0^{(+)}V$ with the states $|\chi_n\rangle$, $|\hat{\chi}_i\rangle$, and $|\vec{k}\rangle$. Numerical codes for the evaluation of these matrix elements in electron-atom and electron-molecule scattering have been developed in recent years.³¹⁻³³ The Schwinger variational principle may of course be replaced by other approximation methods to solve for the scattering in P space. However, approximation methods based on the solution of the Schrödinger equation rather than the LS equation are probably less well suited to incorporate the explicit orthogonalization of the continuum to the discrete state(s).

Nesbet¹⁵ has proposed an approach which is, in principle, equivalent to the present one and based on the orthogonalization of the exact scattering states to the discrete state(s). The full resolvent operator in P space, $\hat{G}^{(\pm)}$, is then given by Eq. (2.31) with $G_0^{(\pm)}$ replaced by $G^{(\pm)}$ everywhere.¹⁵ For practical calculations, especially in electron-molecule scattering, the orthogonalization the free continuum appears to be more manageable than the orthogonalization of the full continuum.

D. Decomposition of the phase shifts

We have shown so far that the projection-operator formalism yields, for a given choice of the projector Q , a unique separation of the full transition matrix into an orthogonality, a direct, and a resonant term. In the special case of scattering from spherically symmetric potentials one often prefers to think in terms of phase shifts of the scattering wave functions in the partial-wave representation. In what follows, we show that the formalism also yields a well-defined separation of the phase shifts into an orthogonality, a direct, and a resonant contribution. For simplicity of notation, we suppress the angular momentum index in what follows. The orthogonality phase shift δ_0 is obtained from the corresponding element of the K matrix (2.29) in the partial-wave representation. The direct scattering phase shift δ_{dir} cannot be obtained from $T_{\text{dir}}^{(F)}$ of Eq. (2.18) according to the usual formula

$$\tan \delta = \frac{-\pi T}{1 - i\pi T} \quad (2.38)$$

since (2.38) would yield a complex phase shift. This is a consequence of the fact that $T_{\text{dir}}^{(F)}$ is computed with orthogonalized asymptotic states $|\vec{k}^{(\pm)}\rangle$ rather than with free asymptotic states $|\vec{k}\rangle$. We may define, however, a new direct scattering transition matrix as

$$T_{\text{dir}} = \langle \hat{\chi}_i | V_{PP} | \hat{\psi}_{\vec{k}}^{(+)} \rangle = S_0^{-1} T_{\text{dir}}^{(F)}, \quad (2.39)$$

where

$$S_0 = e^{2i\delta_0} = 1 - 2\pi i T_0 \quad (2.40)$$

is the orthogonality scattering matrix. Introducing the direct scattering matrix as

$$S_{\text{dir}} = e^{2i\delta_{\text{dir}}} = 1 - 2\pi i T_{\text{dir}} \quad (2.41)$$

it is easily proved that S_{dir} is unitary, i.e., δ_{dir} defined by Eq. (2.41) is real. Introducing, furthermore, the background S matrix, phase shift, and T matrix according to

$$S_{\text{bg}} = S_0 S_{\text{dir}} = e^{2i\delta_{\text{bg}}} = 1 - 2\pi i T_{\text{bg}} \quad (2.42)$$

$$T_{\text{res}} = S_{\text{bg}}^{-1} T_{\text{res}}^{(F)} = \langle \hat{\psi}_k^{(+)} | H_{PQ}(k^2/2 - H_{QQ} - H_{QP} \hat{G}^{(+)} H_{PQ})^{-1} H_{QP} | \hat{\psi}_k^{(+)} \rangle \quad (2.44)$$

and

$$S_{\text{res}} = e^{2i\delta_{\text{res}}} = 1 - 2\pi i T_{\text{res}}. \quad (2.45)$$

As before, it is easy to show that S_{res} is unitary and δ_{res} is real. Finally, it follows that

$$S = S_0 S_{\text{dir}} S_{\text{res}} = S_{\text{bg}} S_{\text{res}}, \quad (2.46)$$

$$\delta = \delta_0 + \delta_{\text{dir}} + \delta_{\text{res}} = \delta_{\text{bg}} + \delta_{\text{res}}, \quad (2.47)$$

where

$$S = e^{2i\delta} = 1 - 2\pi i T \quad (2.48)$$

and T is the full T -matrix element given by Eq. (2.16). We have thus shown that the projection-operator formalism leads not only to a decomposition of the T matrix, but also to a corresponding decomposition of the phase shift (or factorization of the S matrix) into an orthogonality, a direct, and a resonant contribution.

III. PROJECTION-OPERATOR FORMALISM FOR TRUNCATED POTENTIALS

As a pedagogical example of the application of the formalism developed in Sec. II we consider scattering by a finite-rank separable potential of the form

$$V^{(t)} = \sum_{n,m=1}^N |\chi_n\rangle V_{nm} \langle \chi_m|, \quad (3.1a)$$

where

$$V_{nm} = \langle \chi_n | V | \chi_m \rangle \quad (3.1b)$$

and $\{|\vec{r}|\chi_n\rangle\}$ is an arbitrary set of square-integrable basis functions (assumed to be orthonormalized for simplicity). A potential of the form (3.1) is usually called a truncated potential.^{34,35}

As discussed in detail in the Introduction, the idea of the projection-operator formalism in practical applications is to extract one or a few resonances in order to make the remaining background T matrix or phase shift a smooth function of energy in the energy interval considered. Therefore, one would usually choose one or, at most, a few discrete states to span the Q space. The potential of the form (3.1), on the other hand, suggests to choose the Q space such that the potential is fully included in Q space. Defining the projection operator Q as

$$Q = \sum_{n=1}^N |\chi_n\rangle \langle \chi_n| \quad (3.2)$$

we have

it follows

$$T_{\text{bg}} = T_{\text{bg}}^{(F)}, \quad (2.43)$$

where $T_{\text{bg}}^{(F)}$ is defined in Eq. (2.20). To obtain the resonant phase shift δ_{res} we introduce

$$V_{QQ}^{(t)} = V^{(t)}, \quad V_{PQ}^{(t)} = V_{QP}^{(t)} = V_{PP}^{(t)} = 0. \quad (3.3)$$

This property of the potential obviates the solution of the scattering problem in P space, i.e., the $|\hat{\psi}_k^{(\pm)}\rangle$ are identical with the eigenstates $|\vec{k}^{(\pm)}\rangle$ of K_{PP} . In addition, the coupling between the Q and P spaces is solely provided by the kinetic-energy operator K .

Owing to the vanishing of $V_{PP}^{(t)}$, the direct scattering T matrix (2.18) is zero and we have only orthogonality scattering and resonant scattering. In what follows, we shall convince ourselves explicitly that the sum of the orthogonality T matrix $T_0(\vec{k}', \vec{k})$ and the resonant T matrix $T_{\text{res}}^{(F)}(\vec{k}', \vec{k})$ yields the exact T matrix for the potential (3.1).

Taking account of Eq. (3.3), the general expression (2.19) for T_{res} simplifies to

$$T_{\text{res}}^{(F)}(\vec{k}', \vec{k}) = \sum_{n,m=1}^N \langle \vec{k}'^{(-)} | K | \chi_n \rangle (N^{-1})_{nm} \langle \chi_m | K | \vec{k}^{(+)} \rangle, \quad (3.4)$$

where

$$N_{nm} = (k^2/2)\delta_{nm} - H_{nm} - F_{nm}(k), \quad (3.5)$$

$$F_{nm}(k) = \langle \chi_n | K \hat{G}_0^{(+)}(k) K | \chi_m \rangle, \quad (3.6)$$

and $\hat{G}_0^{(+)}$ is given by Eq. (2.31). Making use of the operator identity

$$K \hat{G}_0^{(\pm)} = \hat{G}_0^{(\pm)} K = (k^2/2) \hat{G}_0^{(\pm)} - 1 \quad (3.7)$$

we obtain

$$F_{nm}(k) = (k^2/2)\delta_{nm} - K_{nm} - (\underline{A}^{-1})_{nm}. \quad (3.8)$$

The level-shift matrix $\underline{F}(k)$ is thus independent of the potential for potentials of the form (3.1) and the choice (3.2) for the projector Q . For interpretative purposes, we write $\underline{F}(k)$ in the form

$$\underline{F}(k) = \underline{A}^{-1}(\underline{A} - \underline{\bar{A}})\underline{A}^{-1}, \quad (3.9)$$

where $\underline{\bar{A}}$ is defined as

$$(\underline{\bar{A}})_{nm} = [(\frac{1}{2}k^2 \underline{1} - \underline{K})^{-1}]_{nm} \quad (3.10)$$

and $\underline{A}$ is given by Eq. (2.27) (with outgoing wave boundary condition). $\underline{\bar{A}}$ is the approximate Green's-function matrix constructed by inverting $\frac{1}{2}k^2 \underline{1} - \underline{K}$ in a finite basis of square-integrable functions. $\underline{\bar{A}}$ involves the truncation of the kinetic-energy operator in the basis, in contrast to the

matrix $\underline{A}$, which is obtained by taking matrix elements of the free Green's function. Equation (3.9) shows that the complex level shift $\underline{F}(k)$ just corrects for the truncation of the kinetic-energy operator in H_{QQ} . This establishes the relationship between the Feshbach formalism and the so-called stabilization method.³⁶⁻³⁸ In the stabilization method one diagonalizes the scattering Hamiltonian in a finite set of L^2 functions and tries to identify eigenvalues which are stable with respect to variation of the number of basis functions or variation of a scaling parameter in the basis set. Since H is Hermitian in any L^2 basis, the eigenvalues are constrained to the real energy axis. The level-shift operator (3.9) corrects for the truncation of the kinetic energy in the L^2 basis and shifts the real eigenvalues into the complex plane. As we shall see immediately, it yields the exact poles of the T matrix for the truncated potential (3.1).

Let us continue with the calculation of the resonant T matrix $T_{\text{res}}^{(F)}(\vec{k}', \vec{k})$. Using the explicit expression (2.26) for $|\vec{k}^{(\pm)}\rangle$, we obtain

$$\langle \chi_m | K | \vec{k}^{(\pm)} \rangle = \sum_{m'} (\underline{A}^{-1})_{mm'} \langle \chi_{m'} | \vec{k} \rangle. \quad (3.11)$$

With (3.4), (3.5), (3.8), and (3.11) we arrive at

$$\begin{aligned} T_{\text{res}}^{(F)}(\vec{k}', \vec{k}) \\ = \sum_{n,m=1}^N \langle \vec{k}' | \chi_n \rangle [\underline{A}^{-1}(\underline{1} - \underline{A} \underline{V})^{-1}]_{nm} \langle \chi_m | \vec{k} \rangle. \end{aligned} \quad (3.12)$$

The full T matrix is obtained by adding the orthogonality scattering T matrix of Eq. (2.28). A short calculation gives

$$\begin{aligned} T_0(\vec{k}', \vec{k}) + T_{\text{res}}^{(F)}(\vec{k}', \vec{k}) \\ = \sum_{n,m=1}^N \langle \vec{k}' | \chi_n \rangle [\underline{V}(\underline{1} - \underline{A} \underline{V})^{-1}]_{nm} \langle \chi_m | \vec{k} \rangle. \end{aligned} \quad (3.13)$$

This expression is immediately recognized as the exact T matrix obtained by direct solution of the LS equation for the separable potential (3.1) (see, for example, Refs. 34 and 35). It is thus explicitly shown that the projection-operator formalism (with proper inclusion of the orthogonality scattering) gives the exact solution for any truncated potential when the projector Q spans the range of the potential. In particular, the complex level-shift matrix $\underline{F}(k)$ transforms the (real) eigenvalues of H_{QQ} into the exact poles of the T matrix for the truncated potential.

A special case of these results has recently been obtained by Yamani³⁹ who proved the equivalence of the Feshbach approach with the so-called J -matrix method.³⁵ The J -matrix method is equivalent to the truncation of the (spherically symmetric) potential in a particular L^2 basis, the Laguerre-Slater set.³⁵ The above derivation generalizes the proof of the equivalence of the Feshbach approach with the exact solution for truncated potentials to arbitrary basis sets and nonspherical potentials, illustrating the power of the explicit orthogonalization formulas (2.26), (2.28), and (2.31).

For the calculation of scattering phase shifts the use of truncated potentials has the drawback that it leads, in gen-

eral, to slowly convergent approximation schemes, since the resulting K -matrix or T -matrix elements are not variationally stable.^{31,32} For the calculation of resonances, however, the truncated potential method is still superior to the stabilization method³⁶⁻³⁸ which has found widespread application in electron-molecule scattering.⁴⁰ As discussed above, the stabilization method involves the additional approximation of truncating the kinetic-energy operator in the L^2 basis. While the stabilization method yields only estimates for the real part of the resonance energies, the complex energies obtained as the solutions of Eq. (2.22) approximate directly the complex poles of the T matrix and thus also yield information on the width of the resonances.

IV. EXACT SOLUTION OF MODEL PROBLEMS

In this section I present—presumably for the first time—exact solutions of simple potential scattering problems using the projection-operator approach. This should be the first step in establishing the Feshbach formalism as a mathematical and computational tool for potential scattering. In particular, I want to show that it is feasible and indeed quite simple to solve exactly the direct scattering problem in P space, which was, so far, the obstacle preventing the rigorous application of the Feshbach formalism to potential scattering. Although we restrict ourselves here to very simple models which can be solved in closed form, the general formalism outlined in Sec. II provides a computational scheme directly applicable to more realistic problems (e.g., atom-molecule or electron-molecule scattering).

A. The square-well potential

By far the most common elementary scattering model in nuclear physics is s -wave scattering from the square-well potential, described by the Hamiltonian

$$H = K + V(r), \quad (4.1a)$$

$$K = -\frac{1}{2} d^2/dr^2, \quad (4.1b)$$

$$V(r) = V_0 \Theta(R_0 - r), \quad (4.1c)$$

where $\Theta(x)$ is the Heaviside unit step function. The bound states and scattering states of the Hamiltonian (4.1) are discussed in most textbooks on quantum mechanics. The distribution of the poles of the S -matrix element in the complex momentum plane is also well known from the detailed study of Nussenzweig.⁴¹ In what follows we consider the attractive square-well potential ($V_0 < 0$).

The first step in a projection-operator calculation is to define the set of discrete states which span the Q space. A natural choice for all potentials with a cutoff [i.e., $V(r) = 0$ for $r > R_1$] are the eigenfunctions of a particle in the box with radius R_1

$$\langle r | \chi_j \rangle = \chi_j(r) = \sqrt{2/R_1} \sin(k_j r) \Theta(R_1 - r), \quad (4.2a)$$

$$k_j = j\pi/R_1. \quad (4.2b)$$

These states form an orthonormal set on the interval $[0, R_1]$. For $N \rightarrow \infty$, this set is complete on $[0, R_1]$, i.e.,

$$\sum_{j=1}^{\infty} \chi_j(r) \chi_j(r') = \delta(r-r') \Theta(R_1-r). \quad (4.3)$$

We may now introduce the projector

$$Q(N_1, N_2) = \sum_{\nu=N_1}^{N_2} |\chi_\nu\rangle \langle \chi_\nu|. \quad (4.4)$$

It follows from Eq. (4.3) that $Q(1, \infty)$ is a projection operator in coordinate space, projecting onto the interior of the sphere with radius R_1 . In the following we make the obvious choice $R_1 = R_0$, i.e., the radius of the box defining the discrete states is taken to be equal to the radius of the potential. It is clear that for $R_1 \geq R_0$, $Q(1, \infty)$ spans the range of the potential, i.e.,

$$V_{QQ} = V, \quad V_{PQ} = V_{QP} = V_{PP} = 0 \quad (4.5)$$

for this particular choice of Q .

Having defined Q , we orthogonalize the free s -wave continuum to the set $\{|\chi_\nu\rangle\}$, $N_1 \leq \nu \leq N_2$. The matrix elements of $G_0^{(+)}$ with the basis functions (4.2) are straightforwardly obtained with the use of either the coordinate space representation or the eigenfunction expansion of the free Green's function. The matrix elements are

$$A_{ij} = -2R_0^2 g \frac{(-1)^i + j^2 i \pi^2}{(\beta^2 - i^2 \pi^2)(\beta^2 - j^2 \pi^2)} + 2R_0^2 \frac{\delta_{ij}}{\beta^2 - i^2 \pi^2}, \quad (4.6)$$

where $\beta = kR_0$ is the dimensionless momentum variable and

$$g = \frac{1}{\beta} e^{i\beta} \sin \beta. \quad (4.7)$$

The inverse of the Green's-function matrix (4.6) in the space spanned by the projector (4.4) can be obtained in closed form. To perform this inversion as well as several other matrix inversions required in the ensuing calculations, the following matrix identity is essential:

$$\left(\underline{\Lambda} - \frac{1}{a} \underline{d} \underline{d}^T \right)^{-1} = \underline{\Lambda}^{-1} + \frac{1}{a-b} \underline{\Lambda}^{-1} \underline{d} \underline{d}^T \underline{\Lambda}^{-1}, \quad (4.8a)$$

where $\underline{\Lambda}$ is a diagonal matrix, $\underline{d}$ is a column vector, a is a scalar, and b is given by

$$b = \underline{d}^T \underline{\Lambda}^{-1} \underline{d}. \quad (4.8b)$$

A straightforward evaluation of Eq. (2.29) yields

$$K_0 = \tan \delta_0 = \frac{q \sin^2 \beta}{\beta - q \sin \beta \cos \beta}, \quad (4.9)$$

where

$$q = \sum_{\nu \in Q} \frac{2\nu^2 \pi^2}{\beta^2 - \nu^2 \pi^2}. \quad (4.10)$$

Analogously, Eq. (2.28) yields the orthogonality T -matrix element

$$T_0 = -\frac{1}{\pi} e^{-i\beta} \sin \beta \frac{q}{\beta \cot \beta - q - i\beta}. \quad (4.11)$$

From either (4.9) or (4.11) we obtain the S -matrix element

$$S_0 = e^{-2i\beta} \frac{\beta \cot \beta - q + i\beta}{\beta \cot \beta - q - i\beta}. \quad (4.12)$$

Equations (4.9), (4.11), and (4.12) describe the effect of the orthogonalization of the free s -wave continuum to the arbitrary subset $N_1 \leq \nu \leq N_2$ of box states (4.2).

For the special case of $N_1 = 1$ and $N_2 \rightarrow \infty$ we have $q \rightarrow \infty$ and thus

$$\delta_0 \rightarrow -\beta, \quad (4.13)$$

$$S_0 \rightarrow e^{-2i\beta}. \quad (4.14)$$

$\delta_0 = -\beta$ is just the phase shift obtained for scattering from a hard sphere of radius R_0 .⁴² As expected, orthogonalization of the continuum to the *complete* set of eigenstates of the box forces the orthogonalized continuum wave function to vanish inside the box and thus produces the hard-sphere phase shift.

The orthogonality phase shifts obtained for orthogonalization to $Q(1, N_2)$ with $N_2 = 1-5$ are shown in Fig. 1. When the Q space comprises N discrete states, δ_0 decreases by $N\pi$ radians from $\beta=0$ to $\beta=\infty$. It is obvious from Fig. 1 how the hard-sphere limit (4.13) is approached for $N \rightarrow \infty$. Even for small N , δ_0 is close to the hard-sphere limit (4.13) for small momenta β .

Previous workers^{25,43} have orthogonalized the free continuum to the *exact* bound states of the square-well potential of depth V_0 . The intention was to create a complete and orthonormal set of single-particle states which may be used in an ensuing many-body calculation.^{25,43} I would like to stress that our intention here is different. We orthogonalize the continuum to an essentially arbitrary set of discrete states. The exact bound states of the Hamiltonian under consideration will be obtained after the inclusion of the level shift and analytic continuation into the complex k plane (bound states are given by poles of T or S on the positive imaginary k axis).

The next step in the projection-operator calculation is

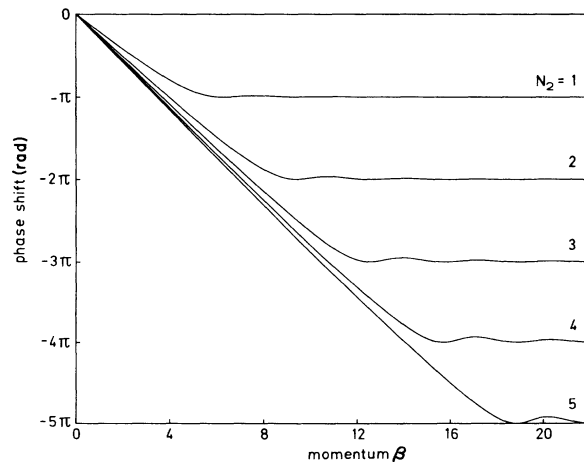


FIG. 1. Orthogonality phase shift δ_0 as a function of the dimensionless momentum variable β , obtained by orthogonalizing the free continuum to the N_2 lowest eigenstates of the particle in a box.

the determination of the direct scattering T matrix $T_{\text{dir}}^{(F)}$. We will evaluate the Schwinger expression (2.34) using the infinite set of functions $\{\chi_n(r)\}$ not included in the Q space as the expansion basis for the potential V_{PP} . Since the total set of functions $\{\chi_i(r)\}$ of Eq. (4.2) is complete over the range of the potential, this will yield the exact solution of the problem.⁴⁴ Due to our choice $R_1 = R_0$, the potential V is diagonal and independent of i in the representation of the $\{\chi_i(r)\}$

$$\langle \chi_i | V | \chi_j \rangle = V_0 \delta_{ij} \quad (4.15)$$

which simplifies the calculations considerably. The matrix elements required in Eq. (2.34) are

$$M_{nm} = V_0 \delta_{nm} - V_0^2 A_{nm} + V_0^2 \sum_{\nu, \mu \in Q} A_{n\nu} (\underline{A}^{-1})_{\nu\mu} A_{\mu m}, \quad (4.16)$$

$$\begin{aligned} \langle \chi_m | V | \hat{k}^{(+)} \rangle \\ = V_0 \langle \chi_m | k \rangle - V_0 \sum_{\nu, \mu \in Q} A_{m\mu} (\underline{A}^{-1})_{\mu\nu} \langle \chi_\nu | k \rangle. \end{aligned} \quad (4.17)$$

The infinite-dimensional matrix $\underline{M}$ (recall that $n, m \notin Q$) can be inverted, again with the use of the identity (4.8). Evaluating Eq. (2.34), we obtain

$$\begin{aligned} T_{\text{dir}}^{(F)} &= \frac{1}{\pi} V_0 R_0^2 \frac{1}{\beta} \sin^2 \beta \frac{2t}{(1-gq)(1-gq+2V_0 R_0^2 gt)} \\ &= \frac{1}{\pi} \beta e^{-2i\beta} \frac{\alpha \cot \alpha - s - \beta \cot \beta + q}{(\beta \cot \beta - q - i\beta)(\alpha \cot \alpha - s - i\beta)}, \end{aligned} \quad (4.18)$$

where

$$\alpha^2 = \beta^2 - 2V_0 R_0^2, \quad (4.19)$$

$$s = \sum_{\nu \in Q} \frac{2\nu^2 \pi^2}{\alpha^2 - \nu^2 \pi^2}, \quad (4.20)$$

$$t = \sum_{n \notin Q} \frac{2n^2 \pi^2}{(\beta^2 - n^2 \pi^2)(\alpha^2 - n^2 \pi^2)}. \quad (4.21)$$

The next step in the calculation is the determination of the complex level-shift matrix $F_{\nu\mu}^{(+)}$. Inserting the Schwinger-type expansion (2.37) for $\hat{G}^{(+)}$ into the definition (2.21) for $F_{\nu\mu}$ we have

$$\begin{aligned} F_{\nu\mu} &= \langle \chi_\nu | K \hat{G}_0^{(+)} K | \chi_\mu \rangle \\ &+ V_0^2 \sum_{n, m \notin Q} \langle \chi_\nu | K \hat{G}_0^{(+)} | \chi_n \rangle \\ &\times (\underline{M}^{-1})_{nm} \langle \chi_m | \hat{G}_0^{(+)} K | \chi_\mu \rangle. \end{aligned} \quad (4.22)$$

Note that in the present example the potential V does not contribute to the coupling between the P and Q spaces, since V is diagonal in the basis $\{\chi_i(r)\}$. A lengthy but straightforward calculation gives

$$F_{\nu\mu} = -\frac{1}{R_0^2} (-1)^{\nu+\mu} \nu \mu \pi^2 \frac{1}{\alpha \cot \alpha - s - i\beta}. \quad (4.23)$$

The width functions $\Gamma_{\nu\mu}(\beta)$ and real level-shift functions $\Delta_{\nu\mu}(\beta)$ are (for real β) obtained as the real and imaginary

parts of $F_{\nu\mu}$:

$$F_{\nu\mu} = \Delta_{\nu\mu} - i/2 \Gamma_{\nu\mu}, \quad (4.24)$$

$$\Delta_{\nu\mu} = -\frac{1}{R_0^2} (-1)^{\nu+\mu} \nu \mu \pi^2 \frac{\alpha \cot \alpha - s}{(\alpha \cot \alpha - s)^2 + \beta^2}, \quad (4.25)$$

$$\Gamma_{\nu\mu} = \frac{2}{R_0^2} (-1)^{\nu+\mu} \nu \mu \pi^2 \frac{\beta}{(\alpha \cot \alpha - s)^2 + \beta^2}. \quad (4.26)$$

The final step of the projection-operator calculation is to obtain the resonant T matrix $T_{\text{res}}^{(F)}$. From Eq. (2.19) we obtain

$$\begin{aligned} T_{\text{res}}^{(F)} &= \sum_{\nu, \mu \in Q} \langle \hat{\psi}_k^{(-)} | K | \chi_\nu \rangle \\ &\times \{[(k^2/2 - V_0)\underline{1} - \underline{K} - \underline{F}]^{-1}\}_{\nu\mu} \\ &\times \langle \chi_\mu | K | \hat{\psi}_k^{(+)} \rangle, \end{aligned} \quad (4.27)$$

where

$$(\underline{K})_{ij} = \langle \chi_i | K | \chi_j \rangle = \frac{1}{2} k_i^2 \delta_{ij} \quad (4.28)$$

and k_i is defined in Eq. (4.2b). To evaluate (4.27), we need matrix elements involving the direct scattering states $|\hat{\psi}_k^{(\pm)}\rangle$. According to Eq. (2.36) these are given by

$$\begin{aligned} \langle \chi_\mu | K | \hat{\psi}_k^{(+)} \rangle \\ = \langle \chi_\mu | K | \hat{k}^{(+)} \rangle \\ + V_0^2 \sum_{n, m \notin Q} \langle \chi_\mu | K \hat{G}_0^{(+)} | \chi_n \rangle (\underline{M}^{-1})_{nm} \langle \chi_m | \hat{k} \rangle. \end{aligned} \quad (4.29)$$

Again using the identity (4.8) to invert the matrices in Eqs. (4.27) and (4.29) we arrive at

$$T_{\text{res}}^{(F)} = \frac{1}{\pi} e^{-2i\beta} \frac{\beta s}{(\alpha \cot \alpha - i\beta)(\alpha \cot \alpha - s - i\beta)}. \quad (4.30)$$

With Eqs. (4.11), (4.18), and (4.30) we finally obtain

$$\begin{aligned} T &= T_0 + T_{\text{dir}}^{(F)} + T_{\text{res}}^{(F)} \\ &= \frac{1}{\pi} e^{-i\beta} \sin \beta \frac{\alpha \cot \alpha - \beta \cot \beta}{\alpha \cot \alpha - i\beta} \end{aligned} \quad (4.31)$$

which is the well-known exact T -matrix element for s -wave scattering from the square-well potential (see, for example, Ref. 41). We have thus explicitly shown that the Feshbach formalism yields the exact solution for arbitrary choices of N_1 and N_2 in the projector $Q(N_1, N_2)$ in Eq. (4.4). Note that it is essential to include the orthogonality T matrix T_0 in the sum (4.31) to obtain the exact result.

The various phase shifts and S -matrix elements can now easily be obtained as outlined in Eqs. (2.39)–(2.48). The resulting expressions for the S -matrix elements are

$$S_{\text{dir}} = \frac{(\beta \cot \beta - q - i\beta)(\alpha \cot \alpha - s + i\beta)}{(\beta \cot \beta - q + i\beta)(\alpha \cot \alpha - s - i\beta)}, \quad (4.32)$$

$$S_{\text{bg}} = e^{-2i\beta} \frac{\alpha \cot \alpha - s + i\beta}{\alpha \cot \alpha - s - i\beta}, \quad (4.33)$$

$$S_{\text{res}} = \frac{(\alpha \cot \alpha + i\beta)(\alpha \cot \alpha - s - i\beta)}{(\alpha \cot \alpha - i\beta)(\alpha \cot \alpha - s + i\beta)}. \quad (4.34)$$

The explicit formulas for the phase shifts are

$$\tan \delta_{\text{dir}} = -\beta \frac{\alpha \cot \alpha - s - \beta \cot \beta + q}{(\alpha \cot \alpha - s)(\beta \cot \beta - q) + \beta^2}, \quad (4.35)$$

$$\tan \delta_{\text{bg}} = -\frac{\alpha \cot \alpha - s - \beta \cot \beta}{\alpha \cot \alpha \cot \beta - s \cot \beta + \beta}, \quad (4.36)$$

$$\tan \delta_{\text{res}} = -\frac{\beta s}{(\alpha \cot \alpha - s)\alpha \cot \alpha + \beta^2}. \quad (4.37)$$

It follows that

$$S = S_0 S_{\text{dir}} S_{\text{res}} = S_{\text{bg}} S_{\text{res}} = e^{-2i\beta} \frac{\alpha \cot \alpha + i\beta}{\alpha \cot \alpha - i\beta}, \quad (4.38)$$

$$\begin{aligned} \delta &= \delta_0 + \delta_{\text{dir}} + \delta_{\text{res}} = \delta_{\text{bg}} + \delta_{\text{res}} \\ &= -\beta + \tan^{-1} \left[\frac{\beta}{\alpha} \tan \alpha \right] + n\pi \end{aligned} \quad (4.39)$$

which are the exact s -wave S -matrix element and phase shift for the square-well potential. This completes the exact solution of the square-well problem within the projection-operator formalism.

Let us begin the discussion of these results by considering the special case $Q = Q(1, \infty)$, where the Q space spans the range of the potential. We then have no direct scattering, only orthogonality scattering and resonant scattering. Since $q \rightarrow \infty$ and $s \rightarrow \infty$ for this choice, it follows immediately from Eqs. (4.18) and (4.32) that $T_{\text{dir}}^{(F)} \rightarrow 0$ and $S_{\text{dir}} \rightarrow 1$. As already discussed above, S_0 becomes the hard-sphere S matrix and from (4.34) it follows that

$$S_{\text{res}} \rightarrow \frac{\alpha \cot \alpha + i\beta}{\alpha \cot \alpha - i\beta} \quad (4.40)$$

which is recognized as the second factor in the exact S matrix (4.38). The appearance of the hard-sphere term $\exp(-2i\beta)$ in the exact S matrix and its importance for the interpretation of resonances has been discussed in detail in the literature.^{45,46} In the present formalism the hard-sphere phase shift (which decreases monotonically with β) arises naturally from the orthogonalization of the continuum to the box states. The resonant phase shift (which rises monotonically with β) results from the coupling of the discrete states in the box to the orthogonalized continuum via the kinetic-energy operator. The situation is thus very transparent in this limit, but a warning is appropriate here. If one computes the matrix element of K between box states $|\chi_\nu\rangle$ and hard-sphere scattering states $|\hat{k}^{(\pm)}\rangle$, one obtains zero for all ν and k , i.e., there is no coupling between the interior and exterior states. The resolution of the paradox is obtained when one computes the coupling element for a finite-dimensional Q space $Q(1, N_2)$, giving

$$\langle \chi_\nu | K | \hat{k}^{(\pm)} \rangle = \frac{1}{\sqrt{\pi\beta}} (-1)^\nu \frac{\nu\pi}{R_0} \frac{\sin\beta}{1-gq}, \quad (4.41)$$

where g and q are defined in Eqs. (4.7) and (4.10). For any fixed index ν , the coupling element vanishes, indeed, for $N_2 \rightarrow \infty$ (since $q \rightarrow \infty$). However, the elements increase with ν ($\nu \leq N_2$) and therefore the limit $N_2 \rightarrow \infty$ has to be taken with care. As shown above, the correct result is obtained when one solves the problem for finite N_2 and then takes the limit $N_2 \rightarrow \infty$.

In practical applications of the Feshbach formalism one

wants to use only a small number of discrete states. The idea is to make the background phase shift a smooth function of energy in a given energy interval. In s -wave scattering from the square-well potential no narrow resonances occur owing to the absence of a barrier. Therefore, we lay the emphasis in the following on the treatment of threshold effects (so-called zero-energy resonances⁴⁷) in the projection-operator formalism. We consider the potential parameters $R_0 = 1$ and $V_0 = -1.2$, for which we have a virtual state close to threshold¹⁶ (this virtual state becomes a bound state for $-V_0 = \pi^2/8 = 1.23\dots$). The exact s -wave phase shift (full line in Fig. 2) shows the steep rise at threshold typical for a virtual state (see, for example, Ref. 42, Sec. 11.2). Choosing the projectors $Q(1,1)$ and $Q(1,2)$, respectively, we obtain from Eqs. (4.36) and (4.37) the phase shifts δ_{bg} and δ_{res} shown in Fig. 2. The orthogonality phase shift δ_0 has already been given in Fig. 1. It is seen that for both choices of the projector the background scattering phase shift δ_{bg} is a smoothly decreasing function of momentum β for small β . The virtual-state effect is fully contained in the resonant phase shift δ_{res} . For $Q(1,1)$, δ_{bg} decreases by π radians and δ_{res} increases by π radians when β varies from 0 to ∞ . For $Q(1,2)$, δ_{bg} decreases by 2π and δ_{res} increases by 2π . Apparently, $Q(1,1)$ projects out the virtual state and one resonance, while $Q(1,2)$ removes the virtual state and two resonances from the background phase shift. The resonances are, however, so broad that they are hardly discernible as those. The example shown in Fig. 2 has also been studied by Nesbet with the use of numerical methods.¹⁶

Figure 3 shows the width function $\Gamma_{11}(\beta)$ and the real level-shift function $\Delta_{11}(\beta)$ [see Eqs. (4.25) and (4.26)] for this example with $Q = Q(1,1)$. Δ_{11} is the Hilbert transform (in the energy variable) of Γ_{11} . Note that $\Gamma_{11}(\beta)$ obeys the Wigner threshold law⁴⁸ $\Gamma_{11} \sim \beta$ for $\beta \rightarrow 0$. These functions play a central role in practical applications of the formalism, for example, in electron-molecule scattering, where the energy dependence of Γ and Δ governs the importance of non-Born-Oppenheimer ef-

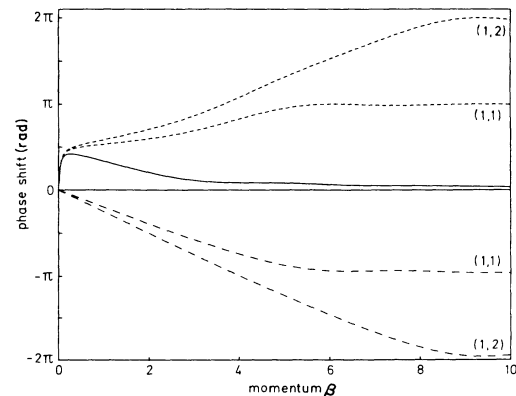


FIG. 2. Background phase shift δ_{bg} (long dashes) and resonant phase shift δ_{res} (short dashes) as functions of momentum β for s -wave scattering from a square-well potential, employing the projector $Q(1,1)$ and $Q(1,2)$, respectively [see Eqs. (4.2) and (4.4)]. In both cases δ_{bg} and δ_{res} add up to the exact phase shift δ (solid line). Steep increase of δ near $\beta=0$ is caused by a virtual state near threshold. Note that the virtual-state effect is fully contained in the resonant phase shift.

fects.^{7,8,11,12} To the knowledge of the author, Γ_{11} and Δ_{11} in Fig. 3 are the first *exact* width and level-shift functions for a potential scattering problem. $F_{11} = \Delta_{11} - i/2\Gamma_{11}$ has the property that it yields the exact poles of the S matrix in the complex k plane as the solutions of the equation

$$k^2/2 - \langle \chi_1 | H | \chi_1 \rangle - F_{11}(k) = 0. \quad (4.42)$$

To understand how the projection-operator formalism achieves the separation of the phase shifts illustrated in Fig. 2, we consider the distribution of the poles of the various S matrices in more detail. According to Eq. (4.38), the poles of the exact S matrix are given by

$$\alpha \cot \alpha - i\beta = 0. \quad (4.43)$$

The solution of this transcendental equation has been discussed in detail by Nussenzveig.⁴¹ According to Eq. (4.33) the poles of the background S matrix are given by

$$\alpha \cot \alpha - s - i\beta = 0, \quad (4.44)$$

where

$$s = \sum_{v=N_1}^{N_2} s_v, \quad s_v = \frac{2v^2\pi^2}{\alpha^2 - v^2\pi^2}. \quad (4.45)$$

s_v has the interesting property that it “annihilates” a pair of solutions of Eq. (4.43). It shifts the pole of the exact S closest to $v\pi$, where this pole is canceled by the corresponding zero of S (poles and zeros of the S matrix are always situated symmetrically to the real axis). Table I shows the solutions of Eq. (4.44) for various choices of the Q space together with the solutions of Eq. (4.43). Taking a single discrete state ($N_1 = N_2 = 1$), the pair of poles ($\pm 4.06, -1.88i$) of the exact S is removed from S_{bg} . With two discrete states ($N_1 = 1, N_2 = 2$), two pairs of poles are removed, and so on. It is interesting that the virtual state (the pole of S on the negative imaginary axis) is not annihilated, but moves down the negative imaginary axis with increasing number of discrete states. The special role played by this pole has been noted previously.^{41,44}

Equation (4.34) shows that the resonant S matrix of the Feshbach formalism contains *all* poles of the exact S matrix. In addition, it possesses *zeros* at the poles of the background S matrix. Table I shows that for the projector $Q(1,1)$ all poles of S_{res} except for the virtual-state pole and the first pair of resonance poles are accompanied by nearby zeros of S_{res} (poles of S_{bg}) which cancel the effect of these poles on the phase shift. For the projector $Q(1,2)$ all poles of S_{res} except the virtual-state pole and the first two pairs of resonance poles are accompanied by nearby

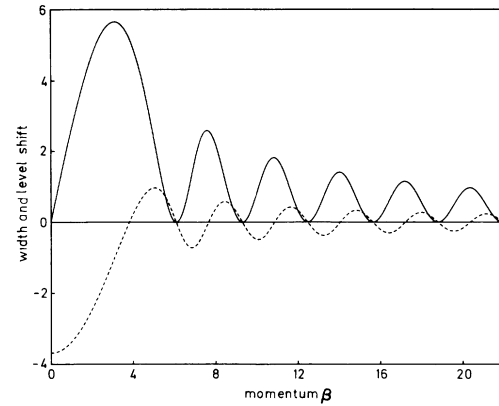


FIG. 3. Exact width (solid line) and level-shift (dashed line) functions for s -wave scattering from the square-well potential, employing the projector $Q(1,1)$, i.e., a single discrete state.

zeros, and so forth. This explains why δ_{res} rises by π for $Q(1,1)$ and by 2π for $Q(1,2)$, as shown in Fig. 2. It shows also how the virtual-state effect is removed from the exact phase shift and incorporated into δ_{res} . The transparent picture emerging from the present exactly soluble model should provide the basis for the better understanding of approximation methods based on the projection-operator formalism.

B. The δ -shell potential

As the simplest example of a potential with a barrier which can support narrow resonances we consider s -wave scattering from the δ -shell potential

$$V(r) = \lambda \delta(r - R_0), \quad (4.46)$$

where $\delta(x)$ is the Dirac δ function and $\lambda > 0$. This problem can be solved by elementary methods.^{45,49,50} The s -wave S -matrix element exhibits an infinite series of resonance poles in the complex k plane situated close to multiples of π/R_0 for $\lambda \rightarrow \infty$.^{49,50} The behavior of the phase shift and its qualitative decomposition into resonance and background contributions has been discussed by Gottfried⁴⁹ and McVoy *et al.*⁴⁵

The solution of the scattering problem using the projection-operator formalism proceeds analogously to Sec. IV A and is, therefore, only briefly sketched. We employ the same Q space as before, i.e., $Q(N_1, N_2)$ as defined in Eqs. (4.2) and (4.4) with the special choice $R_1 = R_0$.

TABLE I. Poles of the background scattering matrix element S_{bg} in the complex momentum plane for s -wave scattering from the square-well potential ($R_0 = 1$, $V_0 = -1.2$) and various projection operators $Q(N_1, N_2)$ defined in Eqs. (4.2) and (4.4). First column ($Q = 0$ means no projection) gives the poles of the exact S -matrix element. Complex poles appear in pairs, situated symmetrically to the imaginary axis. Only the first four pairs of complex poles are given.

$Q=0$	$Q(1,1)$	$Q(1,2)$	$Q(1,3)$
0, -0.0340	0, -15.942	0, -81.859	0, -230.00
$\pm 4.0595, -1.8783$			
$\pm 7.4049, -2.3597$	$\pm 6.6405, -1.7695$		
$\pm 10.646, -2.6820$	$\pm 10.027, -2.3441$	$\pm 9.8278, -1.3539$	
$\pm 13.847, -2.9255$	$\pm 13.332, -2.7090$	$\pm 13.074, -1.8555$	$\pm 13.027, -1.1647$

Note that $V_{QQ} = V_{PQ} = V_{QP} = 0$ for this choice, since the functions $\chi_\nu(r)$ vanish at $r = R_0$. Nevertheless, $Q(N_1, N_2)$ is a useful projector, as will be seen in what follows.

The solution of the scattering problem is greatly simplified by the fact that the potential (4.46) can be written as a separable potential of rank one:

$$V = |f\rangle\lambda\langle f|, \quad (4.47)$$

where $\langle r|f\rangle = \delta(r - R_0)$. Therefore, the direct scattering T matrix is immediately obtained as

$$T_{\text{dir}}^{(F)} = \frac{\langle \hat{k}^{(-)}|f\rangle\langle f|\hat{k}^{(+)}\rangle}{\lambda^{-1} - \langle f|\hat{G}_0^{(+)}|f\rangle} \quad (4.48)$$

and the direct scattering states are

$$|\hat{\psi}_k^{(+)}\rangle = |\hat{k}^{(+)}\rangle + \frac{\hat{G}_0^{(+)}|f\rangle\langle f|\hat{k}^{(+)}\rangle}{\lambda^{-1} - \langle f|\hat{G}_0^{(+)}|f\rangle}. \quad (4.49)$$

$$T_{\text{dir}}^{(F)} = \frac{1}{\pi} \frac{2\lambda R_0 \beta}{(\beta \cot \beta - q - i\beta)(\beta \cot \beta - q + 2\lambda R_0 - i\beta)}, \quad (4.52)$$

$$S_{\text{dir}} = \frac{(\beta \cot \beta - q - i\beta)(\beta \cot \beta - q + 2\lambda R_0 + i\beta)}{(\beta \cot \beta - q + i\beta)(\beta \cot \beta - q + 2\lambda R_0 - i\beta)}, \quad (4.53)$$

$$\tan \delta_{\text{dir}} = \frac{-2\lambda R_0 \beta}{(\beta \cot \beta - q + 2\lambda R_0)(\beta \cot \beta - q) + \beta^2}, \quad (4.54)$$

$$T_{\text{bg}}^{(F)} = \frac{1}{\pi} e^{-i\beta} \sin \beta \frac{2\lambda R_0 - q}{\beta \cot \beta - q + 2\lambda R_0 - i\beta}, \quad (4.55)$$

$$S_{\text{bg}} = e^{-2i\beta} \frac{\beta \cot \beta - q + 2\lambda R_0 + i\beta}{\beta \cot \beta - q + 2\lambda R_0 - i\beta}, \quad (4.56)$$

$$\tan \delta_{\text{bg}} = -\frac{\tan \beta (2\lambda R_0 - q)}{\beta \cot \beta - q + 2\lambda R_0 + \beta \tan \beta}, \quad (4.57)$$

$$T_{\text{res}}^{(F)} = \frac{1}{\pi} e^{-2i\beta} \frac{q\beta}{(\beta \cot \beta - q + 2\lambda R_0 - i\beta)(\beta \cot \beta + 2\lambda R_0 - i\beta)}, \quad (4.58)$$

$$S_{\text{res}} = \frac{(\beta \cot \beta - q + 2\lambda R_0 - i\beta)(\beta \cot \beta + 2\lambda R_0 + i\beta)}{(\beta \cot \beta - q + 2\lambda R_0 + i\beta)(\beta \cot \beta + 2\lambda R_0 - i\beta)}, \quad (4.59)$$

$$\tan \delta_{\text{res}} = \frac{-q\beta}{(\beta \cot \beta - q + 2\lambda R_0)(\beta \cot \beta + 2\lambda R_0) + \beta^2}. \quad (4.60)$$

The quantity q is defined in Eq. (4.10). From these results we obtain the exact T matrix, S matrix, and phase shift according to

$$\begin{aligned} T &= T_0 + T_{\text{dir}}^{(F)} + T_{\text{res}}^{(F)} \\ &= \frac{1}{\pi} e^{-i\beta} \sin \beta \frac{2\lambda R_0}{\beta \cot \beta + 2\lambda R_0 - i\beta}, \end{aligned} \quad (4.61)$$

$$S = S_0 S_{\text{dir}} S_{\text{res}} = e^{-2i\beta} \frac{\beta \cot \beta + 2\lambda R_0 + i\beta}{\beta \cot \beta + 2\lambda R_0 - i\beta}, \quad (4.62)$$

$$\begin{aligned} \tan \delta &= (\delta_0 + \delta_{\text{dir}} + \delta_{\text{res}}) \\ &= \frac{-2\lambda R_0 \sin^2 \beta}{2\lambda R_0 \sin \beta \cos \beta + \beta}. \end{aligned} \quad (4.63)$$

The complex level shift becomes

$$\begin{aligned} F_{\nu\mu}(k) &= \langle \chi_\nu | K \hat{G}_0^{(+)} | \chi_\mu \rangle \\ &\quad - \frac{\langle \chi_\nu | K \hat{G}_0^{(+)} | f \rangle \langle f | \hat{G}_0^{(+)} | \chi_\mu \rangle}{\lambda^{-1} - \langle f | \hat{G}_0^{(+)} | f \rangle} \end{aligned} \quad (4.50)$$

and the resonant T matrix is finally obtained as

$$\begin{aligned} T_{\text{res}}^{(F)} &= \sum_{\nu, \mu \in Q} \langle \hat{\psi}_k^{(-)} | K | \chi_\nu \rangle \\ &\quad \times [(\frac{1}{2}k^2 \mathbb{1} - \underline{K} - \underline{F})^{-1}]_{\nu\mu} \langle \chi_\mu | K | \hat{\psi}_k^{(+)} \rangle. \end{aligned} \quad (4.51)$$

The orthogonality T matrix, S matrix, and phase shift are, of course, the same as in Sec. IV A [see Eqs. (4.9)–(4.12)]. In complete analogy to the square-well problem, we obtain the following for the direct, the background, and the resonant scattering:

Finally, we give the width and real level-shift functions

$$\Gamma_{\nu\mu} = \frac{2}{R_0^2} (-1)^{\nu+\mu} \nu\mu \pi^2 \frac{\beta}{(\beta \cot \beta - q + 2\lambda R_0)^2 + \beta^2}, \quad (4.64)$$

$$\begin{aligned} \Delta_{\nu\mu} &= -\frac{1}{R_0^2} (-1)^{\nu+\mu} \nu\mu \pi^2 \\ &\quad \times \frac{\beta \cot \beta - q + 2\lambda R_0}{(\beta \cot \beta - q + 2\lambda R_0)^2 + \beta^2}. \end{aligned} \quad (4.65)$$

Let us illustrate the results obtained for the example $\lambda = 10$, i.e., a fairly opaque barrier leading to narrow resonances. Figure 4 shows the resonant and background

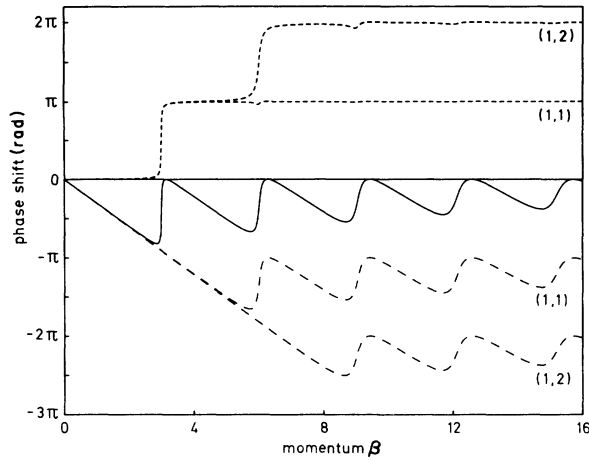


FIG. 4. Background phase shift δ_{bg} (long dashes) and the resonant phase shift δ_{res} (short dashes) as a function of momentum β for s -wave scattering from the δ -shell potential, employing the projectors $Q(1,1)$ and $Q(1,2)$, respectively [see Eqs. (4.2) and (4.4)]. In both cases δ_{bg} and δ_{res} add up to the exact phase shift δ (solid line). It is seen that $Q(1,1)$ removes the lowest resonance from δ_{bg} , while $Q(1,2)$ removes the two lowest resonances.

phase shifts obtained from Eqs. (4.57) and (4.60) together with the full phase shift (4.63) for the choices $Q(1,1)$ and $Q(1,2)$. It is seen that $Q(1,1)$ indeed projects out the first resonance near $\beta = \pi$, i.e., δ_{bg} is a smooth and monotonously decreasing function of β up to $\beta \approx 2\pi$, while δ_{res} shows a sharp increase by π radians at $\beta \approx \pi$. $Q(1,2)$ projects out two resonances, i.e., δ_{bg} is a smooth and monotonously decreasing function up to $\beta \approx 3\pi$, while δ_{res} exhibits two resonant increases by π rad at $\beta \approx \pi$ and $\beta \approx 2\pi$. In this way, we can remove an arbitrary number of resonances from the full phase shift δ . It is also possible to project out selectively higher resonances, e.g., only the second resonance near $\beta \approx 2\pi$. Obviously, $|\chi_\nu\rangle\langle\chi_\nu|$ serves as a projector onto the ν th resonance.

To understand how the projection-operator formalism achieves the decomposition of the phase shift shown in Fig. 4, we consider again the poles of the various S matrices in the complex β plane. According to Eq. (4.62) the exact poles are given by

$$\beta \cot \beta + 2\lambda R_0 - i\beta = 0 \quad (4.66)$$

while the poles of the background S matrix are obtained from [see Eq. (4.56)]

$$\beta \cot \beta - q + 2\lambda R_0 - i\beta = 0. \quad (4.67)$$

As we have seen before,

$$q = \sum_{\nu=N_1}^{N_2} \frac{2\nu^2\pi^2}{\beta^2 - \nu^2\pi^2}$$

has the effect that it eliminates $N_2 - N_1 + 1$ pairs of solutions of Eq. (4.66). This is illustrated in Table II, where the poles of S are shown together with the poles of S_{bg} for $N_1 = 1$ and $N_2 = 1-3$. It is seen that a successive number of poles of S_{bg} disappear as N_2 increases. The remaining poles differ only slightly from the poles of the full S matrix. Equation (4.59) shows that S_{res} possesses the poles of the full S matrix, and that it has zeros at the poles of S_{bg} . It follows that all but $N_2 - N_1 + 1$ poles of S_{res} are approximately canceled by nearby zeros. Therefore, δ_{res} rises by $(N_2 - N_1 + 1)\pi$ rad when β varies from 0 to ∞ , as illustrated in Fig. 4. The small dips in $\delta_{res}(\beta)$ at the positions of the higher resonances visible in Fig. 4 are a consequence of the nonperfect cancellation of the poles of S_{res} .

To conclude this section, we show the exact width and level-shift functions $\Gamma_{11}(\beta)$ and $\Delta_{11}(\beta)$ for $Q(1,1)$ [Fig. 5(a)] and $Q(1,2)$ [Fig. 5(b)]. In the latter case, the width and the level shift are 2×2 matrices, of which we show only the (1,1) element. The β dependence of the (1,2) and (2,2) elements is very similar to the (1,1) element. The width and level shift functions vary strongly with β since the poles of the background S matrix are also poles of the complex level shift $\Delta_{\nu\mu} - \frac{1}{2}i\Gamma_{\nu\mu}$. As more and more discrete states are included in the Q space, more and more resonances in the background scattering disappear and $\Gamma_{\nu\mu}(\beta)$ and $\Delta_{\nu\mu}(\beta)$ become smooth functions. It is noteworthy that $\Gamma_{\nu\mu}$ becomes approximately linear in β and $\Delta_{\nu\mu}$ approximately constant, i.e., both are given by the lowest term in their threshold expansion to a good approximation once the poles are removed. Figure 5 may serve as a reminder that the width and real level-shift functions cannot *a priori* be assumed to be smooth and structureless functions of the momentum or the energy.

V. CONCLUSIONS

A straightforward scheme to perform projection-operator calculations for potential scattering has been developed. It is based on the explicit orthogonalization of the free continuum to a set of discrete states spanning the Q space and the ensuing solution of the scattering problem in P space with the use of separable expansions of the potential. It has been shown that the formalism yields a well-defined decomposition of the transition matrix into an orthogonality, a direct, and a resonant term. Each of

TABLE II. Poles of the background scattering matrix element S_{bg} in the complex momentum plane for s -wave scattering from the δ -shell potential ($R_0 = 1$, $\lambda = 10$) and various projection operators $Q(N_1, N_2)$ defined in Eqs. (4.2) and (4.4). First column gives the poles of the exact S -matrix element. Only the first five pairs of poles are given.

$Q=0$	$Q(1,1)$	$Q(1,2)$	$Q(1,3)$
$\pm 2.9958, -0.0205$			
$\pm 6.0109, -0.0744$	$\pm 6.0027, -0.0795$		
$\pm 9.0533, -0.1457$	$\pm 9.0499, -0.1487$	$\pm 9.0266, -0.1734$	
$\pm 12.120, -0.2215$	$\pm 12.119, -0.2236$	$\pm 12.110, -0.2352$	$\pm 12.075, -0.2966$
$\pm 15.207, -0.2950$	$\pm 15.206, -0.2966$	$\pm 15.201, -0.3040$	$\pm 15.187, -0.3284$

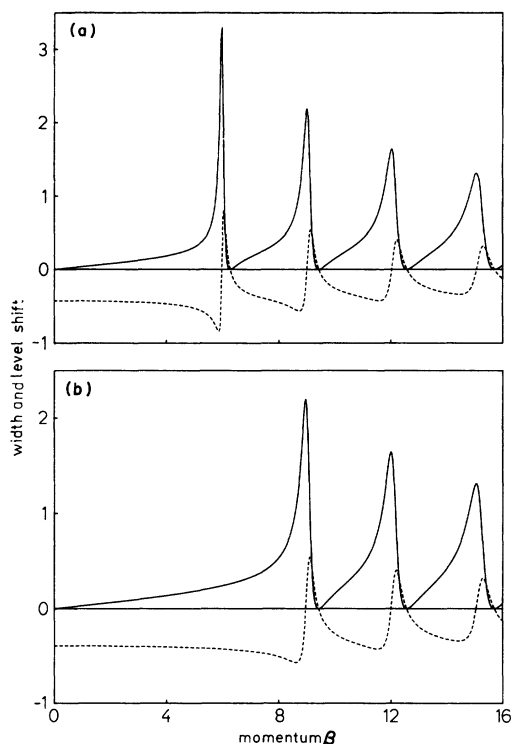


FIG. 5. Exact width (solid line) and level-shift (dashed line) functions for s -wave scattering from the δ -shell potential. (a) $\Gamma_{11}(\beta)$ and $\Delta_{11}(\beta)$ for the case of a single discrete state, (b) $\Gamma_{11}(\beta)$ and $\Delta_{11}(\beta)$ for the case of two discrete states. (1,2) and (2,2) elements of Γ and Δ (not shown) exhibit a similar dependence on β .

the three terms can be calculated explicitly. The method is well-suited to handle scattering from nonspherical potentials, for example, electron-molecule or atom-molecule scattering.

The method has been illustrated for two simple exactly soluble model problems, namely, s -wave scattering from the square-well and the δ -shell potential. Both problems have been solved exactly in the projection-operator formalism employing an arbitrary number of eigenstates of the particle in a box to span the Q space. It has been shown that the formalism yields a separation of the full phase shift into a smooth background term and a resonant term which may vary rapidly with energy. The analytic structure of the T and S matrices appearing in the projection-operator formalism has been discussed. A finite number of poles of the full S matrix do not appear in the background scattering matrix S_{bg} , defined as the product of the orthogonality and the direct S matrices. The remaining poles of S_{bg} are situated close to the poles of the full S . The resonant scattering matrix S_{res} , on the other hand, possesses the same poles as the exact S matrix. All but a finite number of these poles are approximately canceled by nearby zeros (which are the poles of S_{bg}). For the square-well potential it was shown how virtual-state effects are removed from S_{bg} in the projection-operator formalism. The virtual-state pole is not annihilated, but retreats from the origin as more and more discrete states are included in the Q space.

The relevance of these results for practical calculations lies in the application of the BO approximation to the electron-molecule scattering problem. Once resonances and virtual and bound states close to threshold are removed, the background scattering can be treated in the BO approximation (the so-called adiabatic-nuclei approximation, Ref. 17). The resonant T matrix of the Feshbach formalism can be calculated beyond the BO approximation by taking account of the nonlocality of the optical potential governing the nuclear motion.^{7,8,12,51}

A question which deserves some discussion is the appropriate choice of the discrete state(s) defining the Q space. The formalism as such is applicable to any choice of the projector Q . The problem is to ensure that a chosen discrete state "removes" a particular resonance, i.e., makes the background phase shift δ_{bg} a smooth function of energy in the vicinity of the resonance and transfers the resonance into the resonant phase shift δ_{res} . In particular, the projection procedure should not introduce spurious resonances into δ_{bg} at energies where the full phase shift δ is nonresonant. In the two examples studied in this paper we employed a set of discrete states (the eigenstates of a particle in a box) and found *a posteriori* that these states are appropriate, i.e., each discrete state included in the Q space removed exactly one pair of resonance poles from the background S matrix. For the δ -shell potential it is intuitively clear that the resonances are closely related to the discrete states of the particle in a box of the same radius. For a shallow square-well potential this connection is not as obvious; it is therefore interesting to see that the projection-operator formalism indeed establishes a one-to-one correspondence between the discrete box states and the poles of the S matrix.

In more realistic problems, in particular, in scattering from nonspherical potentials, the appropriate choice of the discrete state(s) is less clear. There is a significant amount of literature devoted to the question of how to approximate shape resonances by L^2 wave functions and how to determine these wave functions variationally. The most common method in practical applications is the so-called stabilization method.^{37,38,40} Here the Rayleigh-Ritz minimization of the energy (which, in the absence of bound states, would lead to a wave function describing a projectile with zero kinetic energy at infinite distance from the target) is replaced by the concept of stabilization of the energy. In practice, it is required that an eigenvalue, obtained by diagonalizing H in an L^2 basis, should be stable either with respect to the extension of the basis set or with respect to a scaling of the basis functions.^{37,38} Stable eigenvalues are believed to approximate resonances. Discrete states obtained by this method have been successfully employed by Hazi and co-workers to calculate the width function $\Gamma(E)$ of the shape resonance in $e\text{-N}_2$ and $e\text{-F}_2$ scattering.¹² Another approach is to minimize the variance of the Hamiltonian, $\langle H^2 \rangle - \langle H \rangle^2$, with the use of L^2 test functions.⁵²⁻⁵⁴ This method was successfully used by Read to calculate autoionizing electronic states of the He atom.⁵² It should be stressed that one needs only a crude estimate of the discrete state when applying the Feshbach formalism, because the real level shift $\Delta(E)$ corrects for the error, yielding the exact resonance energy (provided the orthogonality scattering and the direct scattering problems are solved exactly).

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- ¹H. Feshbach, Ann. Phys. (N.Y.) **5**, 357 (1958).
- ²H. Feshbach, Ann. Phys. (N.Y.) **19**, 287 (1962).
- ³See, for example, R. H. Lemmer, Rep. Prog. Phys. **24**, 131 (1966).
- ⁴See, for example, T. F. O'Malley and S. Geltman, Phys. Rev. **137**, A1344 (1965).
- ⁵J. L. Dehmer and D. Dill, in *Electron-Molecule and Photon-Molecule Collisions*, edited by T. N. Rescigno, V. McKoy, and B. Schneider (Plenum, New York, 1979), p. 225.
- ⁶For reviews, see H. Massey, *Negative Ions*, 3rd ed. (Cambridge University Press, London, 1976); J. N. Bardsley and M. A. Biondi, Adv. At. Mol. Phys. **6**, 1 (1971).
- ⁷J. C. Y. Chen, J. Chem. Phys. **40**, 3507 (1964); **40**, 3513 (1964); Phys. Rev. **148**, 66 (1966).
- ⁸T. F. O'Malley, Phys. Rev. **150**, 14 (1966).
- ⁹A. Herzenberg, J. Phys. B **1**, 548 (1968).
- ¹⁰H. Nakamura, J. Phys. Soc. Japan **31**, 574 (1971).
- ¹¹W. Domcke, J. Phys. B **14**, 4889 (1981); **16**, 359 (1983).
- ¹²A. U. Hazi, T. N. Rescigno, and M. Kurilla, Phys. Rev. A **23**, 1089 (1981); A. U. Hazi, A. E. Orel, and T. N. Rescigno, Phys. Rev. Lett. **46**, 918 (1981).
- ¹³A. U. Hazi, J. Phys. B **11**, L259 (1978); and in Ref. 5, p. 281.
- ¹⁴P. W. Langhoff, Int. J. Quant. Chem. Symp. **8**, 347 (1974).
- ¹⁵R. K. Nesbet, *Variational Methods in Electron-Atom Scattering Theory* (Plenum, New York, 1980), Sec. 3.2.
- ¹⁶R. K. Nesbet, Phys. Rev. A **24**, 1184 (1981).
- ¹⁷See, for example, N. F. Lane, Rev. Mod. Phys. **52**, 29 (1980).
- ¹⁸T. F. O'Malley, Adv. At. Mol. Phys. **7**, 223 (1971).
- ¹⁹W. Lichten, Phys. Rev. **131**, 229 (1963).
- ²⁰H. C. Longuet-Higgins, Adv. Spectrosc. **2**, 429 (1961).
- ²¹J. Schwinger, Phys. Rev. **72**, 742 (1947); B. A. Lippmann and J. Schwinger, Phys. Rev. **79**, 469 (1950).
- ²²L. Fonda and R. G. Newton, Ann. Phys. (N.Y.) **10**, 490 (1960); R. G. Newton, *Scattering Theory of Waves and Particles* (McGraw-Hill, New York, 1966), Sec. 16.5.
- ²³M. M. Stigl and M. W. Kirson, Nucl. Phys. **A137**, 289 (1969).
- ²⁴G. E. Juras, J. E. Monahan, C. M. Shakin, and R. M. Thaler, Phys. Rev. B **5**, 4000 (1972).
- ²⁵R. R. Scheerbaum, C. M. Shakin, and R. M. Thaler, Ann. Phys. (N.Y.) **76**, 333 (1973).
- ²⁶D. C. Mattis, Phys. Lett. **63A**, 184 (1977).
- ²⁷R. Sugar and R. Blencenbecler, Phys. Rev. **136**, B472 (1964).
- ²⁸S. K. Adhikari and I. H. Sloan, Phys. Rev. C **11**, (1973) 1133.
- ²⁹A. L. Zubarev, Sov. J. Part. Nucl. **9**, 188 (1978) [Fiz. Elem. Chastits. At. Yadra **9**, 453 (1978)].
- ³⁰K. Takatsuka and V. McKoy, Phys. Rev. A **23**, 2352 (1981).
- ³¹D. K. Watson and V. McKoy, Phys. Rev. A **20**, 1474 (1979).
- ³²M. Berman and U. Kaldor, J. Phys. B **14**, 3993 (1981).
- ³³R. R. Lucchese and V. McKoy, Phys. Rev. A **21**, 112 (1980).
- ³⁴T. N. Rescigno, C. W. McCurdy, and V. McKoy, Phys. Rev. A **10**, 2240 (1974); **11**, 825 (1975).
- ³⁵E. J. Heller and H. A. Yamani, Phys. Rev. A **9**, 1201 (1974).
- ³⁶E. Holøien and J. Midtdal, J. Chem. Phys. **45**, 2209 (1966).
- ³⁷A. U. Hazi and H. S. Taylor, Phys. Rev. A **1**, 1109 (1970).
- ³⁸M. F. Fels and A. U. Hazi, Phys. Rev. A **4**, 662 (1971).
- ³⁹H. A. Yamani, J. Math. Phys. **23**, 83 (1982).
- ⁴⁰See, for example, H. S. Taylor, J. K. Williams, and I. Eliezer, J. Chem. Phys. **47**, 2165 (1967); E. Goldstein, G. A. Segal, and R. W. Wetmore, J. Chem. Phys. **68**, 271 (1978).
- ⁴¹H. M. Nussenzveig, Nucl. Phys. **11**, 499 (1959).
- ⁴²R. G. Newton, *Scattering Theory of Waves and Particles* (McGraw-Hill, New York, 1966), Sec. 14.2.
- ⁴³B. H. Brandow, Ann. Phys. (N.Y.) **57**, 214 (1970).
- ⁴⁴W. Domcke, J. Phys. B **15**, 2675 (1982).
- ⁴⁵K. W. McVoy, L. Heller, and M. Bolsterli, Rev. Mod. Phys. **39**, 245 (1967).
- ⁴⁶H.-D. Meyer and K. T. Tang, Z. Phys. A **279**, 349 (1976).
- ⁴⁷J. R. Taylor, *Scattering Theory* (Wiley, New York, 1972), Chap. 12.
- ⁴⁸E. P. Wigner, Phys. Rev. **73**, 1002 (1948).
- ⁴⁹K. Gottfried, *Quantum Mechanics* (Benjamin, New York, 1966), Chap. 15.
- ⁵⁰H. M. Nussenzveig, Nuovo Cimento **20**, 694 (1961).
- ⁵¹M. Berman, H. Estrada, L. S. Cederbaum, and W. Domcke, Phys. Rev. A **28**, 1363 (1983).
- ⁵²F. H. Read, Chem. Phys. Lett. **12**, 549 (1972).
- ⁵³P. Froelich and E. Brändas, Phys. Rev. A **12**, 1 (1975).
- ⁵⁴A. Macías and A. Riera, J. Phys. B **14**, 955 (1981).